

## **FisherChain PH: Transparent Seafood Marketplace**

### **Introduction**

Community Impact FisherChain PH contributes to economic empowerment by increasing fishermen's income and reducing dependency on exploitative intermediaries. It promotes fair trade practices, enhances trust within the seafood industry, and encourages digital financial inclusion. By strengthening transparency and accountability, the system supports sustainable economic growth in Southern Leyte and the Philippines.

### **Problem Statement**

Fishermen in Southern Leyte frequently experience unfair pricing due to reliance on middlemen who control market access and distribution channels. Because transactions are rarely documented in a secure and transparent system, fishermen often lack proof of agreed prices and payment terms. This results in reduced income, delayed payments, and limited bargaining power. Additionally, buyers have little visibility into product origin, pricing breakdown, and supply chain tracking. These challenges weaken trust and hinder economic growth within the local fishing sector.

### **Technology Explanation**

Blockchain is a secure and decentralized digital ledger that records transactions permanently. Once recorded, data cannot be altered or deleted. Every transaction is verified by the network, making it transparent and tamper-proof.

It works by:

Recording transactions in blocks Linking blocks together in a chain Distributing copies of the ledger across multiple nodes Ensuring security through cryptographic validation This makes fraud and data manipulation nearly impossible.

### **Application Features**

Catch Registration System – Enables fishermen to record daily catch details including type, quantity, and suggested price.

- Direct Digital Payments – Allows buyers to transfer payments instantly through mobile wallets

- Supply Chain Tracking – Records product movement from fisherman to buyer using blockchain verification.

- Price Transparency Dashboard – Displays real-time pricing data to ensure fair trade

- practices. •Fraud-Proof Records – Stores permanent transaction logs that cannot be altered or deleted.

### **Target Users**

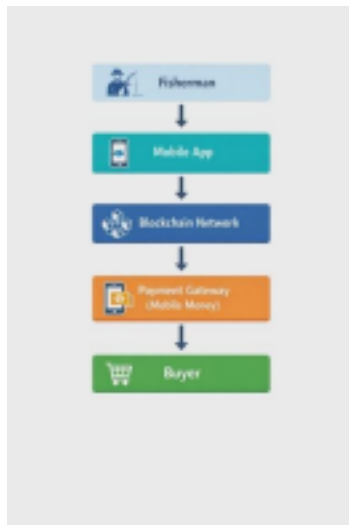
The primary users include fishermen in Southern Leyte, seafood buyers, restaurants, market vendors, and fishing cooperatives. The platform also benefits local government agriculture and

fisheries offices by providing transparent data records that support policy planning and economic monitoring.

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### **System Architecture**



### **Enhancement Requirement**

Why Blockchain + Mobile Money is Effective Blockchain ensures transparent and tamper proof transaction records. It prevents corruption, price manipulation, and hidden fees. Mobile money ensures fast, secure, and cashless payment processing. It allows fishermen to receive money instantly without waiting for physical cash. Together, they eliminate middlemen corruption, ensure fair pricing, and create a secure digital seafood marketplace.

### **Reflection**

Why did you choose this technology?

I chose Blockchain and Mobile Money because they provide transparency and security in financial transactions. Many fishermen in Southern Leyte are exploited by middlemen who manipulate prices. Blockchain ensures that every transaction is permanently recorded and cannot be altered. This prevents fraud and promotes trust between sellers and buyers. Mobile Money allows fast, secure, and cashless payments directly to the fishermen. Combining these technologies creates a fair and efficient seafood marketplace. It also encourages the adoption of digital technology in rural communities. Blockchain provides real-time supply chain tracking,

which improves accountability. Mobile Money makes payments accessible even in remote areas with smartphones. I believe this combination addresses both financial and operational issues in the local fishing industry. The technology is also scalable and can be applied to other agricultural or local trade systems. Overall, it empowers fishermen and improves economic sustainability.

What challenges might occur in developing this?

One challenge is limited internet access in coastal communities, which may affect real-time data recording. Some fishermen may not be familiar with using mobile apps or digital payments. Blockchain implementation requires technical expertise and a secure network infrastructure. Integrating Mobile Money with blockchain can be complex and needs careful programming. Resistance from middlemen who benefit from the current system may also occur. Ensuring data privacy and compliance with local regulations is another challenge. The initial cost of developing and deploying IoT or mobile-enabled systems can be high. Training users to adopt the system is necessary to make it effective. Maintenance and troubleshooting of the application can be challenging in remote areas. Collecting accurate data for blockchain entries is critical. There may also be occasional software bugs or system downtime. Overcoming these challenges requires planning, community engagement, and technical skill development.

Is it realistic? Why or why not?

Yes, it is realistic, but it requires proper planning and support. Mobile Money is already widely used in the Philippines, which makes payment integration feasible. Smartphones and internet coverage are increasingly accessible, even in rural areas. Blockchain networks can be implemented with cloud-based solutions to reduce infrastructure costs. Gradual adoption can allow fishermen to learn the system over time. Partnerships with local government units and cooperatives can provide support and training. Initial funding might be needed to cover development and deployment costs. Technical support teams can monitor and maintain the system. Farmers and fishermen are already familiar with using apps for communication and trading. The transparency and security benefits justify the effort required to implement the system. Over time, this system can become a sustainable solution for fair trade. The concept can also scale to other provinces or industries with minor adjustments.

What skills must you improve to build this in the future?

I need to improve my skills in blockchain development to ensure secure and tamper-proof transaction recording. Mobile app development skills are necessary to build a user-friendly interface for fishermen. Knowledge of Mobile Money APIs and integration methods is also essential. I should enhance my database management skills to store and retrieve transaction data efficiently. Cybersecurity skills are important to protect sensitive financial information. Cloud computing knowledge will help in hosting the blockchain network reliably. I also need skills in IoT device integration for real-time catch registration. User experience design skills will ensure that fishermen can easily use the app. Project management skills are needed to coordinate development, testing, and deployment. Data analysis skills will help in generating pricing insights for users. I should also improve communication and training skills for community onboarding. Continuous learning in emerging technologies will keep me updated and

capable of building this system successfully.